

Tailwind CSS with React – Complete Beginner-Friendly Notes

What is Tailwind CSS?

Tailwind CSS is a modern **utility-first CSS framework**. Instead of writing custom CSS classes, you compose styles by applying pre-defined utility classes directly to your HTML or JSX elements.

Example:

Instead of writing this:

```
CSS                                                                    Copy Edit

.card { background-color: red; padding: 1rem; border-radius: 8px; text-align: center; }
```

And then:

```
html                                                                    Copy Edit

<div class="card">Hello</div>
```

With Tailwind:

```
html                                                                    Copy Edit

<div class="bg-red-500 p-4 rounded-lg text-center">Hello</div>
```

- ✓ Faster styling
- ✓ Fully responsive
- ✓ Highly customizable
- ✓ No need to write CSS unless you want to extend it

Why use Tailwind with React?

- Works great with React's component-based structure.
- Keeps styles in the same file as the component (JSX).
- Encourages consistent design using utility classes.

Tailwind CSS Installation in React

This guide assumes you're starting with a basic React project (via Create React App or Vite).

Step 1: Install Tailwind

In your React project directory:

```
bash
```



Copy



Edit

```
npm install -D tailwindcss postcss autoprefixer npx tailwindcss init
```

✓ This creates a `tailwind.config.js` file.

Step 2: Configure Tailwind

Edit `tailwind.config.js` like this:

```
js
```



Copy



Edit

```
/** @type {import('tailwindcss').Config} */ module.exports = { content: ["../src/**/*.{js,jsx,ts,tsx}"], // Important: tells Tailwind where to scan for classes theme: { extend: {}, }, plugins: [], }
```

 This makes Tailwind scan all `.js`, `.jsx`, `.ts`, and `.tsx` files in `src/` to generate styles only for the classes you use (tree-shaking for CSS).

Step 3: Create Tailwind CSS File

In `src/index.css`, add the following Tailwind directives:

```
CSS
```



Copy



Edit

```
@tailwind base; @tailwind components; @tailwind utilities;
```

These import Tailwind's base styles, components, and utility classes.

⚙️ Step 4: Compile Tailwind CSS

Run this CLI command to generate the final CSS:

```
bash
```



Copy



Edit

```
npx tailwindcss -i ./src/index.css -o ./src/output.css --watch
```

✂️ This compiles Tailwind classes into `output.css` and watches for any changes in real time.



Step 5: Import the Generated CSS

In `src/index.js` (React app entry point), import the compiled Tailwind styles:

```
js
```



Copy



Edit

```
import './output.css'; // Import compiled styles import React from 'react'; import ReactDOM from 'react-dom/client'; import App from './App'; const root = ReactDOM.createRoot(document.getElementById('root')); root.render(<App />);
```

Now Tailwind classes are globally available across your components!



Tailwind in Action – Example Component

Let's create a component using Tailwind classes:

Design.js

```
jsx
```



Copy



Edit

```
import React from 'react'; function Design() { return ( <div className="grid grid-cols-1 md:grid-cols-4 gap-5 p-4"> <div className="bg-red-600 h-60 text-white flex justify-center items-center">hi</div> <div className="bg-amber-500 h-60 text-white flex justify-center items-center">hey</div> <div className="bg-lime-700 h-60 text-white flex justify-center items-center">hello</div> <div className="bg-indigo-700 h-60 text-white flex justify-center items-center">oi</div> </div> ); } export default Design;
```

Explanation of Classes:

- `grid grid-cols-1 md:grid-cols-4` : One column on small screens, four on medium and up.
- `gap-5` : spacing between items.
- `p-4` : padding inside container.
- `bg-*` : background color classes.
- `h-60` : fixed height.
- `text-white` : white text.
- `flex justify-center items-center` : centers content both horizontally and vertically.

Bonus Tips

Add Tailwind as a script in `package.json` :

json



Copy



Edit

```
"scripts": { "tailwind": "tailwindcss -i ./src/index.css -o ./src/output.css --watch" }
```

Now just run:

bash



Copy



Edit

```
npm run tailwind
```

Tailwind Dev Tools

Install this VS Code extension:

Tailwind CSS IntelliSense

- Auto-suggestions
- Class previews
- Error checking

 Summary

Step	Description
1	Install Tailwind
2	Configure <code>tailwind.config.js</code>
3	Create and update <code>index.css</code>
4	Compile using Tailwind CLI
5	Import <code>output.css</code> in <code>index.js</code>
6	Start using Tailwind classes in JSX

